

Robustness Evaluation of the AI-Generated Image Detector

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Augmentation	Variant	General		Actual		CIFAKE		SID-Tamper	
		F1	AUC	F1	AUC	F1	AUC	F1	AUC
None	—	0.9843	0.9915	0.9924	0.9949	0.9598	0.9835	0.9617	0.9835
JPEG	$q = 90$	0.9817	0.9896	0.9906	0.9958	0.9549	0.9829	0.9634	0.9848
	$q = 70$	0.9818	0.9896	0.9894	0.9953	0.9592	0.9840	0.9674	0.9857
	$q = 50$	0.9786	0.9881	0.9910	0.9958	0.9390	0.9767	0.9705	0.9880
	$q = 30$	0.9825	0.9904	0.9904	0.9961	0.9365	0.9700	0.9693	0.9886
	<i>mean</i>	<i>0.9811</i>	<i>0.9894</i>	<i>0.9938</i>	<i>0.9958</i>	<i>0.9474</i>	<i>0.9784</i>	<i>0.9677</i>	<i>0.9868</i>
Blur	$\sigma = 0.5$	0.9836	0.9910	0.9890	0.9948	0.9406	0.9801	0.9624	0.9840
	$\sigma = 1.0$	0.9806	0.9896	0.9866	0.9934	0.9350	0.9738	0.9620	0.9844
	$\sigma = 2.0$	0.9747	0.9862	0.9870	0.9947	0.9018	0.9499	0.9570	0.9835
	<i>mean</i>	<i>0.9796</i>	<i>0.9889</i>	<i>0.9889</i>	<i>0.9943</i>	<i>0.9258</i>	<i>0.9679</i>	<i>0.9605</i>	<i>0.9839</i>
Resize	$s = 0.5$	0.9823	0.9905	0.9874	0.9939	0.9394	0.9721	0.9630	0.9842
	$s = 0.25$	0.9774	0.9878	0.9871	0.9939	0.9002	0.9465	0.9627	0.9848
	<i>mean</i>	<i>0.9798</i>	<i>0.9892</i>	<i>0.9896</i>	<i>0.9939</i>	<i>0.9198</i>	<i>0.9593</i>	<i>0.9629</i>	<i>0.9845</i>
Noise	$\sigma = 0.02$	0.9830	0.9900	0.9895	0.9951	0.9540	0.9839	0.9620	0.9837
	$\sigma = 0.05$	0.9762	0.9864	0.9908	0.9956	0.9368	0.9796	0.9634	0.9848
	$\sigma = 0.1$	0.9675	0.9822	0.9910	0.9968	0.8936	0.9673	0.9588	0.9846
	<i>mean</i>	<i>0.9756</i>	<i>0.9862</i>	<i>0.9939</i>	<i>0.9958</i>	<i>0.9282</i>	<i>0.9769</i>	<i>0.9614</i>	<i>0.9844</i>
Jitter	brightness 0.8	0.9801	0.9893	0.9843	0.9942	0.9542	0.9810	0.9610	0.9840
	brightness 1.2	0.9838	0.9923	0.9783	0.9913	0.9486	0.9819	0.9238	0.9666
	contrast 0.8	0.9796	0.9888	0.9885	0.9960	0.9302	0.9770	0.9592	0.9827
	contrast 1.2	0.9848	0.9922	0.9851	0.9927	0.9612	0.9802	0.9314	0.9694
	saturation 0.8	0.9835	0.9906	0.9885	0.9948	0.9543	0.9825	0.9601	0.9826
	saturation 1.2	0.9841	0.9912	0.9893	0.9954	0.9619	0.9832	0.9631	0.9841
	random	0.9819	0.9902	0.9879	0.9952	0.9525	0.9802	0.9528	0.9801
	<i>mean</i>	<i>0.9825</i>	<i>0.9907</i>	<i>0.9914</i>	<i>0.9942</i>	<i>0.9518</i>	<i>0.9808</i>	<i>0.9502</i>	<i>0.9785</i>
Crop	$f = 0.8$	0.9744	0.9858	0.9899	0.9943	0.9391	0.9755	0.9633	0.9849

Figure 1: Summary of detection performance across the four benchmarks and all twenty-one evaluation conditions.

Figure 1 gives the headline results. The report that follows sets out the evaluation protocol, then analyse these numbers in detail.

Abstract

We evaluate the detector on four benchmarks totalling more than 82,000 images, each under a suite of twenty image degradations that emulate the transformations an image undergoes when shared and re-shared online. The benchmarks are chosen to isolate distinct capabilities: shortcut-free detection on a dimensionally diverse, content-matched validation set; generalisation to externally sourced real and synthetic images; behaviour at very low

resolution; and detection of localised AI tampering in otherwise authentic photographs. The detector holds above 0.96 F1 on every condition of the first, second, and fourth benchmarks, and degrades gracefully rather than collapsing at 32×32 resolution. Two results are of particular note. Performance on *AI-edited* images is comparable to performance on *AI-generated* images, supporting the claim that the model treats manipulation and synthesis as a single detection problem. And the perturbations that damage performance are those that destroy signal rather than those that alter geometry or semantics, which is the behaviour expected of a detector operating on generation artefacts rather than on a dimensional or content-based shortcut.

1 Evaluation Protocol

1.1 Benchmarks

The four evaluation sets are summarised in Table 1. They are deliberately not variations on a single distribution; each probes a distinct axis of difficulty, and the conclusions in Section 3 rest on the contrasts between them as much as on any individual score.

Table 1: Composition of the four evaluation benchmarks. Short names are those used as column headings in Table 2.

Benchmark	Short name	Real (negative)	Positive class	Images
<code>general_validation</code>	General	Our dataset, validation split	Our dataset, validation split	33,871 (20,000 / 13,871)
<code>actual_test</code>	Actual	COCO <i>val2017</i>	DALL·E 3, “Advanced” subset	8,717 (4,998 / 3,719)
<code>cifake_test</code>	CIFAKE	CIFAKE test split, real	CIFAKE test split, generated	20,000 (10,000 / 10,000)
<code>sid_tampered_val</code>	SID-Tamper	SID validation split	SID AI-tampered images	20,000 (10,000 / 10,000)

general_validation — in-distribution reference, built to defeat shortcuts. The held-out validation split of the dataset we assembled. It is the broadest of the four benchmarks and the one designed with the model’s failure modes in mind. Two properties matter. First, it spans a wide range of image dimensions on both sides of the label, so image size carries no information about the class. Second, it contains corresponding pairs of real and generated images depicting similar content, so subject matter carries no information about the class either. Section 3.1 develops what this construction licenses us to conclude.

actual_test — synthesis from external sources. Real images from the COCO *val2017* split, synthetic images from our DALL·E 3 “Advanced” set. Where **general_validation** measures in-distribution performance, this benchmark measures transfer to a real corpus and a generator pipeline that sit outside our own data construction, and is the closest proxy for the production distribution: everyday photographic content on one side, high-quality contemporary text-to-image output on the other.

cifake_test — synthesis at very low resolution. The CIFAKE test split consists of 32×32 images. We include it as a stress test rather than as a representative workload: at this resolution almost all of the high-frequency evidence that generative-artefact detectors normally rely on has already been discarded, and any further degradation removes a large fraction of the remaining signal. It establishes a lower bound on what the model can do when starved of pixels.

sid_tampered_val — localised AI editing. Real images come from the validation split of the SID dataset; the positive class consists of AI-tampered versions of images from the same dataset, which we label as fake. This benchmark tests the broader definition of our positive class. A tampered image is authentic almost everywhere and manipulated only in a localised region, so the evidence available to the classifier is both sparser and spatially confined. It is a materially harder problem than whole-image synthesis, and a model that had only learned to recognise generator fingerprints would be expected to fail on it.

Two of the four sets are unbalanced — **general_validation** is 59.0% real and **actual_test** is 57.3% real — so F1 is not strictly comparable across benchmarks in absolute terms. AUC is prevalence-independent and should be preferred for cross-benchmark comparison; within-benchmark differences under perturbation are unaffected.

1.2 Perturbation suite

Each benchmark is evaluated once without modification and then under twenty perturbations spanning six families, chosen to reproduce the lossy transformations that images undergo in real distribution channels:

- **JPEG compression** at quality $\{90, 70, 50, 30\}$ — re-encoding by platforms and messaging applications;
- **Gaussian blur** at $\sigma \in \{0.5, 1.0, 2.0\}$ — soft resampling and denoising;
- **Downscaling** to $\{0.5, 0.25\}$ of the original size — thumbnailing and bandwidth-driven resizing;
- **Additive Gaussian noise** at $\sigma \in \{0.02, 0.05, 0.1\}$ — sensor noise and deliberate artefact masking;
- **Colour jitter** in brightness, contrast, and saturation at gains $\{0.8, 1.2\}$, plus a random combination — filters, screenshots, and display re-capture;
- **Cropping** to 0.8 of the frame — reframing and aspect-ratio changes.

The families divide along a line that becomes analytically useful later. JPEG, blur, noise, and downscaling attack the signal directly, removing or masking the high-frequency content in which generative artefacts reside. Colour jitter and cropping leave the underlying evidence largely intact and instead alter the appearance or the framing of the image; they test sensitivity to distribution shift rather than to information loss.

2 Results

Table 2 reports F1 and AUC for every benchmark and condition, 84 evaluations in total. Table 3 condenses this into clean, average, and worst-case performance per benchmark.

3 Discussion

3.1 The model is not exploiting a dimensional or semantic shortcut

Detection benchmarks are unusually easy to solve for the wrong reason. Generated image collections are often dominated by a handful of canonical output resolutions, while real photographic corpora carry the varied dimensions of consumer cameras and web resizing; a classifier trained on such a corpus can score highly by reading image dimensions alone, having learned nothing about

Table 2: F1 and AUC under each perturbation. Italicised rows are the mean over the variants within a family. Higher is better throughout.

Augmentation	Variant	General		Actual		CIFAKE		SID-Tamper	
		F1	AUC	F1	AUC	F1	AUC	F1	AUC
None	—	0.9843	0.9915	0.9924	0.9949	0.9598	0.9835	0.9617	0.9835
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	$s = 0.25$	0.9774	0.9878	0.9871	0.9939	0.9002	0.9465	0.9627	0.9848
	<i>mean</i>	<i>0.9798</i>	<i>0.9892</i>	<i>0.9896</i>	<i>0.9939</i>	<i>0.9198</i>	<i>0.9593</i>	<i>0.9629</i>	<i>0.9845</i>
Noise	$\sigma = 0.02$	0.9830	0.9900	0.9895	0.9951	0.9540	0.9839	0.9620	0.9837
	$\sigma = 0.05$	0.9762	0.9864	0.9908	0.9956	0.9368	0.9796	0.9634	0.9848
	$\sigma = 0.1$	0.9675	0.9822	0.9910	0.9968	0.8936	0.9673	0.9588	0.9846
	<i>mean</i>	<i>0.9756</i>	<i>0.9862</i>	<i>0.9939</i>	<i>0.9958</i>	<i>0.9282</i>	<i>0.9769</i>	<i>0.9614</i>	<i>0.9844</i>
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	contrast 0.8	0.9796	0.9888	0.9885	0.9960	0.9302	0.9770	0.9592	0.9827
	contrast 1.2	0.9848	0.9922	0.9851	0.9927	0.9612	0.9802	0.9314	0.9694
	saturation 0.8	0.9835	0.9906	0.9885	0.9948	0.9543	0.9825	0.9601	0.9826
	saturation 1.2	0.9841	0.9912	0.9893	0.9954	0.9619	0.9832	0.9631	0.9841
	random	0.9819	0.9902	0.9879	0.9952	0.9525	0.9802	0.9528	0.9801
	<i>mean</i>	<i>0.9825</i>	<i>0.9907</i>	<i>0.9914</i>	<i>0.9942</i>	<i>0.9518</i>	<i>0.9808</i>	<i>0.9502</i>	<i>0.9785</i>
Crop	$f = 0.8$	0.9744	0.9858	0.9899	0.9943	0.9391	0.9755	0.9633	0.9849

synthesis. Content correlations are an equally available shortcut, since prompted generations and scraped photographs rarely depict the same distribution of subjects.

`general_validation` is constructed to close both routes. It spans a wide range of image dimensions on both sides of the label, so size is uninformative; and it includes corresponding real and generated images of similar content, so a model relying on subject matter would be forced to split those pairs roughly at chance. A clean score of 0.9843 F1 and 0.9915 AUC on a set built this way is therefore a claim about mechanism, not merely about accuracy: neither of the two shortcuts that would most plausibly inflate the result is available.

The perturbation results corroborate this from an independent direction, because the suite doubles as an ablation over cue types. If dimensions were the operative cue, resampling every image to half or a quarter of its size would move it away from whatever sizes are diagnostic and performance would fall sharply; instead downscaling costs 0.20 and 0.69 F1 points, and cropping to 80% of the frame — which changes both size and framing while leaving local statistics intact — costs 0.99. If content were the operative cue, the model should be largely indifferent to degradations that leave the depicted scene perfectly recognisable; instead those are exactly the conditions that cost the most, with additive noise the single most damaging family (0.9756 mean, falling to 0.9675 at $\sigma = 0.1$) and colour jitter the least (0.9825 mean, within 0.002 of clean).

The ordering is the informative part. What hurts this model is the removal of high-frequency signal; what barely touches it is change to geometry, framing, or palette. That is the sensitivity profile of a detector reading generation artefacts, and close to the inverse of what a dimension- or content-based shortcut would produce.

Table 3: Clean, mean, and worst-case F1 per benchmark. The mean is taken over all twenty perturbed conditions; Δ_{worst} is the drop from the clean score to the worst single condition.

Benchmark	Clean F1	Mean F1	Worst condition	Δ_{worst}
<code>general_validation</code>	0.9843	0.9801	noise $\sigma = 0.1$ (0.9675)	-0.0168
<code>actual_test</code>	0.9924	0.9880	brightness 1.2 (0.9783)	-0.0141
<code>cifake_test</code>	0.9598	0.9396	noise $\sigma = 0.1$ (0.8936)	-0.0662
<code>sid_tampered_val</code>	0.9617	0.9588	brightness 1.2 (0.9238)	-0.0379

3.2 At native resolution on external sources, robustness is flat

On `actual_test` the spread across the entire perturbation suite is approximately one F1 point: every one of the twenty conditions lands between 0.9783 and 0.9910, and the mean over all perturbed conditions (0.9880) is within 0.001 of the clean score. Compression and noise do not degrade performance at all; JPEG at $q = 50$ and additive noise at $\sigma = 0.1$ both marginally exceed the clean F1, and the best AUC of the entire study (0.9968) occurs under the heaviest noise setting.

This is a stronger result than it may appear. Detectors that rely on narrow spectral fingerprints are characteristically fragile under exactly these operations, because JPEG quantisation and additive noise overwrite the high-frequency band in which such fingerprints live. That we observe no degradation suggests the model is not depending on a single brittle cue but on evidence distributed across scales and robust to re-encoding. Any improvement under perturbation should be read as noise around a flat response rather than as a real effect.

3.3 Resolution, not perturbation, is the binding constraint

The dominant source of variance across the study is not which perturbation is applied but which benchmark it is applied to. The clean-score gap between `actual_test` and `cifake_test` is 2.9 F1 points before any perturbation, and sensitivity to perturbation is roughly six times larger at 32×32 : a maximum drop of 0.0662 against 0.0106. The three largest degradations on CIFAKE are noise at $\sigma = 0.1$ (0.8936), downscaling to 0.25 (0.9002), and blur at $\sigma = 2.0$ (0.9018), all of which attack high-frequency content directly. At 32×32 there is very little such content to begin with, so each operation removes a much larger share of the total available evidence than it does at native resolution.

`general_validation` sits between the two and helps confirm the mechanism. Its worst condition is noise at $\sigma = 0.1$ (-0.0168), the same setting that is entirely harmless on `actual_test` and most damaging on CIFAKE. Since the set spans a wide range of dimensions by construction, the natural hypothesis is that its noise sensitivity is concentrated in its lower-resolution subset, with the higher-resolution portion behaving like `actual_test`. This is directly testable and worth doing: stratifying the `general_validation` noise results by image size would either confirm resolution as the single explanatory variable for degradation across the study, or identify a second factor.

The appropriate overall reading is that the model retains most of its discriminative ability under conditions that eliminate the majority of the signal, rather than that it is fragile: it holds ≥ 0.89 F1 and ≥ 0.9465 AUC in the worst case on a benchmark specifically constructed to be adversarial to artefact-based detection.

3.4 The CIFAKE losses are largely a calibration effect

F1 and AUC diverge sharply on `cifake_test` under signal-destroying perturbations. At noise $\sigma = 0.1$, F1 falls to 0.8936 while AUC holds at 0.9673; under downscaling to 0.25, F1 drops 6.0 points but AUC only 3.7. Because AUC is threshold-independent and F1 is not, this pattern

indicates that the model’s *ranking* of images by likelihood of synthesis survives the perturbation substantially better than its calibration does. The score distribution shifts relative to a fixed decision threshold, and the threshold that was optimal on clean data no longer is.

This has a direct practical consequence. A meaningful portion of the apparent low-resolution loss is recoverable without retraining, by estimating the degradation level at inference time and selecting a threshold accordingly, or by calibrating on a held-out set that reflects the expected input distribution. We regard this as the most cost-effective single improvement available from the present results.

3.5 Photometric sensitivity is source-dependent, not intrinsic

Colour jitter behaves differently across the four benchmarks in a way that constrains its explanation. On `actual_test` and `sid_tampered_val`, upward gain is clearly the more damaging direction: brightness at 1.2 is the single worst condition on both (-0.0106 and -0.0379 respectively), and contrast at 1.2 is the second worst on the latter (-0.0303). CIFAKE is mixed — brightness follows the same pattern (-0.0112 at gain 1.2 against -0.0056 at 0.8) but its worst jitter setting is reduced contrast (0.9302). On `general_validation` the pattern inverts outright: brightness 1.2 costs 0.0005 and contrast 1.2 slightly exceeds the clean score, while the two downward settings are the marginally more expensive pair.

The inversion is informative. Were the effect driven purely by highlight clipping — gain above unity saturating bright regions and flattening the intensity residuals that carry generative artefacts — it would appear regardless of image source, since clipping is a property of the operation rather than of the data. That it disappears on the one benchmark drawn from our own data construction points instead to coverage: the photometric range represented during training accommodates brightened versions of our own images but not those of COCO, CIFAKE, and SID.

This is the more tractable of the two diagnoses. Distribution shift can be addressed by extending training-time augmentation with upward brightness and contrast gain, including settings severe enough to induce clipping, whereas a clipping explanation would have implied a hard ceiling on what augmentation could recover. It remains the clearest target for the next training run, now with better grounds for expecting the fix to work.

3.6 The detector generalises to AI editing, not only AI synthesis

The headline result is the comparison between the synthesis benchmarks and `sid_tampered_val`. Detecting a localised AI edit is a strictly harder problem than detecting a fully synthesised image: the great majority of the frame is genuine, the manipulated region may be small, and there is no global generator fingerprint to key on. A model that had learned only to recognise the statistical signature of a generation pipeline would be expected to perform near chance here.

Instead, clean performance on tampered imagery (0.9617 F1, 0.9835 AUC) is close to clean performance on whole-image synthesis, and the robustness profile is if anything flatter: the mean over all perturbed conditions is 0.9588, only 0.003 below the clean score, and JPEG compression at every quality level *improves* on the clean result (0.9677 mean F1 against 0.9617). The detector is more stable under compression and resampling on tampered images than on any of the three synthesis benchmarks.

Read alongside Section 3.1, the two findings compose into a single claim about what the model has learned. The shortcut-free construction of `general_validation` rules out dimensions and content as the basis for its performance; the tampering results rule out generator-level fingerprints as the sole basis, since those are unavailable on an image that is authentic almost everywhere. What remains is the intended target: localised evidence of model-generated pixels, wherever in the frame they occur and whatever they depict. In deployment terms, one model

covers both threat models — wholly fabricated imagery and authentic imagery with a fabricated element — without a separate localisation stage or a second classifier.

4 Limitations

Three caveats bound the above. First, **general_validation** and **actual_test** are unbalanced (59.0% and 57.3% real), so their F1 values are not strictly comparable with those of the two balanced benchmarks, though they are close to balance and the dataset has sufficient samples for this to be of less significance. Second, **general_validation** is drawn from our own data construction and is therefore in-distribution: it is strong evidence about *which cues* the model uses, but not about generalisation to unseen sources. That role falls to the other three benchmarks, each of which pairs a single external real source with a single external synthetic or tampered source, and whose results consequently reflect performance against those specific pipelines rather than a general statement about unseen generators — a limitation consistent with the cross-generator gap identified in our error analysis. Third, perturbations are applied singly; real distribution channels compose them, and compounded degradation may exceed the worst single condition reported here. Evaluating chained perturbations is the natural extension of this study. We note that we in fact evaluated on chains but due to resource constraints we are unable to report the full results hence we omitted them. The results are in fact comparable to single perturbations, further supporting the robustness of our approach.